

**Thank you** for purchasing our Ariel Voice Generation asset.

This document will show you how to use the asset in Unreal Engine.

If you have any question, contact us at : [contact@xandimmersion.com](mailto:contact@xandimmersion.com)

## Table of contents

|                                                                            |           |
|----------------------------------------------------------------------------|-----------|
| <b>1. Prerequisites</b>                                                    | <b>3</b>  |
| <b>2. How to install the Ariel plugin in your project</b>                  | <b>4</b>  |
| 2.1. Through the Unreal Marketplace                                        | 4         |
| 2.2. Directly                                                              | 4         |
| <b>3. How to use the Ariel Plugin demo Actor</b>                           | <b>5</b>  |
| <b>4. How to use the Ariel plugin in Blueprints (Runtime &amp; Editor)</b> | <b>6</b>  |
| 4.1. Ariel Text-To-Speech                                                  | 6         |
| 4.2. Download audio as bytes array                                         | 9         |
| 4.3. Download audio and save it to a file                                  | 10        |
| <b>5. How to use the Ariel plugin in Editor (with graphical interface)</b> | <b>11</b> |
| <b>6. Tags</b>                                                             | <b>12</b> |
| 6.1. Pause tag                                                             | 12        |
| <b>7. Speaker list</b>                                                     | <b>12</b> |
| 7.1. Premium voices                                                        | 12        |
| 7.2. Standard voices                                                       | 14        |

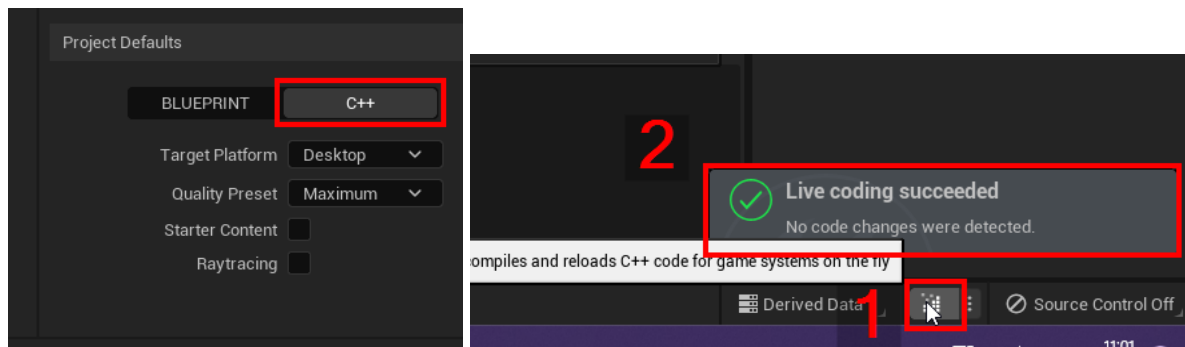
## 1. Prerequisites

There is two different version of the plugin:

- Compiled plugin (for Unreal engine 5.1.1 and 5.0.3)
- Uncompiled plugin \*

\* If you download the uncompiled version of the plugin or if you have another version of Unreal (i.e.: UE 5.2), you must be able to compile C++ code for Unreal Engine. To do so, you need the Visual Studio build tools for Unreal Engine installed as well as the .NET SDK ! You can find more information on how to set up Visual studio for Unreal engine [here](#).

To ensure that you have all build tools installed, try to create a new C++ Unreal Project and make sure that the Editor starts correctly. Once the editor is open, click on the compile button on the bottom right corner and ensure that the compilation succeeded:



## 2. How to install the Ariel plugin in your project

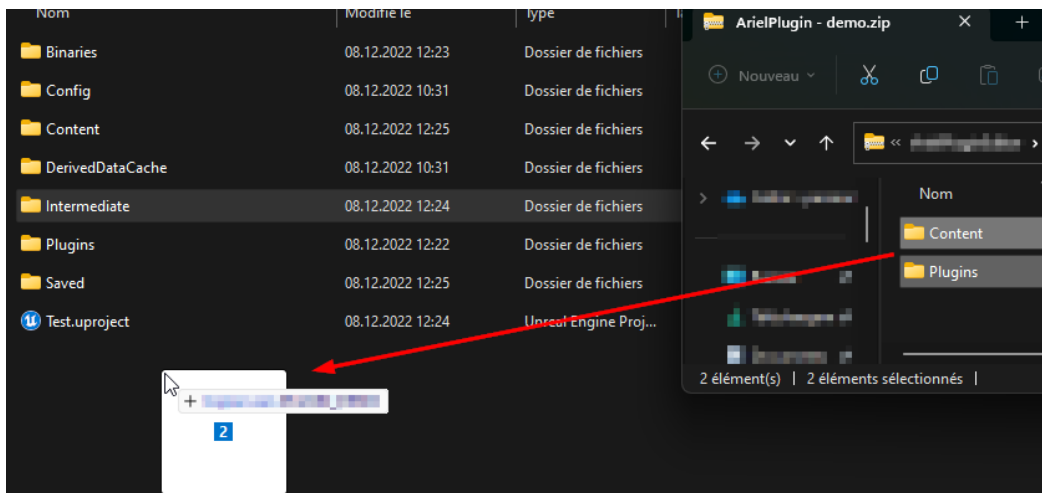
### 2.1. Through the Unreal Marketplace

If you bought the plugin through the Unreal Marketplace, you can simply click on the “Add to project” button. The list of all plugins is located at the bottom of the “Library” tab.

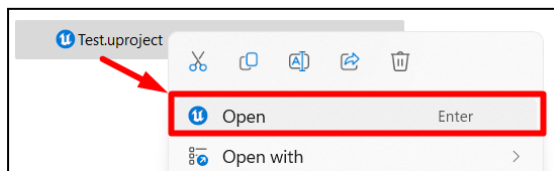
### 2.2. Directly

**First, close the unreal engine project where you want Ariel to be installed !**

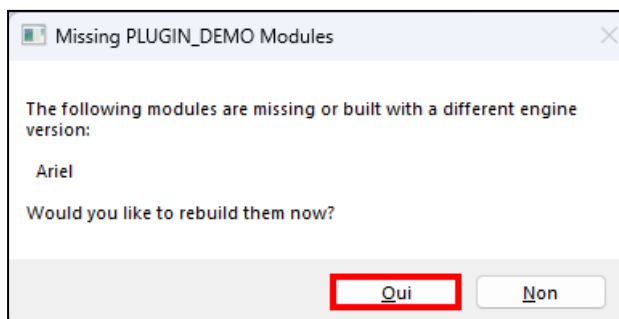
Extract the content from the zip archive and paste it to your base project directory:



Then, reopen your Unreal project (Right click on <Project\_name>.uproject ⇒ Open):



*Optional:* If the “Missing Modules” Window appears, choose “Yes” \*:

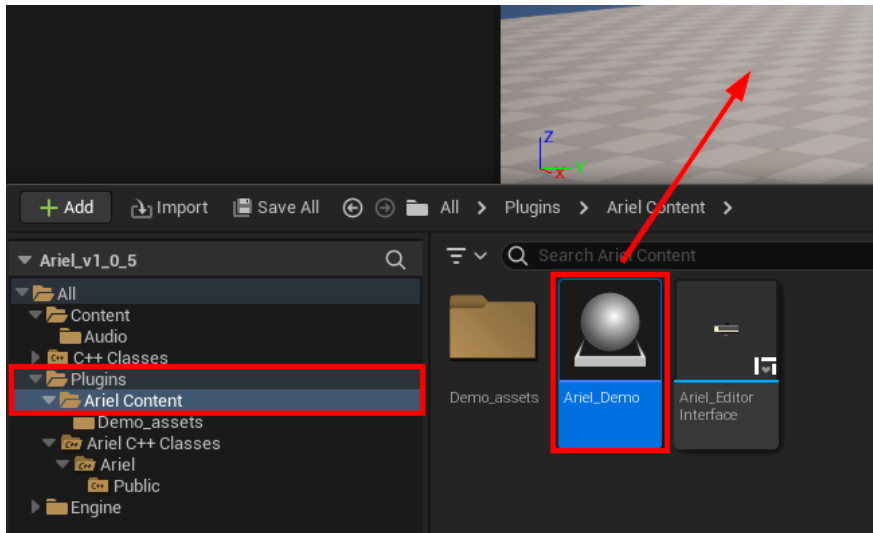


\* This message appears if there is no compiled version of the plugin or if the compilation was made with a different version of the Unreal Engine. If the message “[...] Could not be compiled. Try rebuilding from source manually” appears, please make sure that you have the build tools and SDK installed, as documented in the prerequisites.

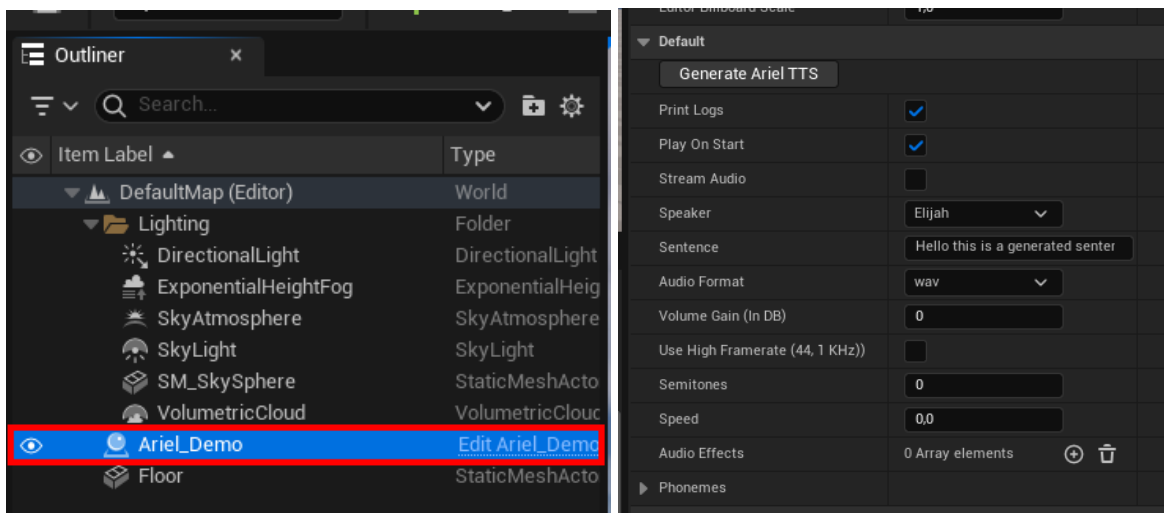
### 3. How to use the Ariel Plugin demo Actor

You can find a fully operational Blueprint Actor in the content folder of the plugin. **In order to see plugin Content, please make sure that you have checked “Show Plugin Content” in the Content Browser settings.** The demo actor calls the Ariel API to generate a sentence with the specified parameters when the level is started.

First, drag and drop the Actor located in the plugin content files > demo in your level:



Then, you can select the actor from the level outline and edit the settings in the details: \*



\* By default, the Ariel sentence is generated when the level starts. You can call it again at runtime by pressing the “Call Ariel” button.

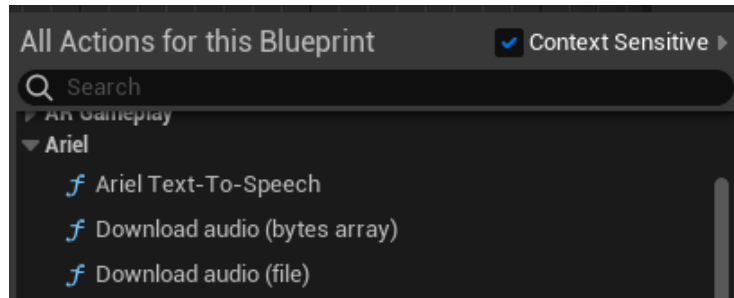
You can find more information about every parameter in the chapter [Ariel Text-To-Speech](#) below. In addition, there is two additional parameters:

**Play on start:** Indicate if the speech is generated when the game starts.

**Stream audio:** Indicate if the generated audio should be streamed or downloaded first.

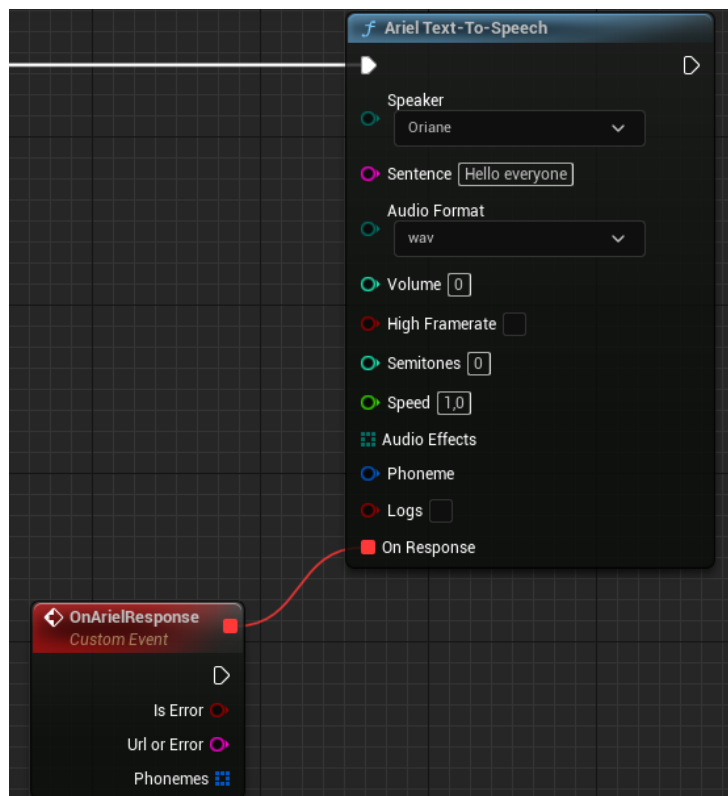
## 4. How to use the Ariel plugin in Blueprints (Runtime & Editor)

The plugin have 3 functions that can be used directly in Blueprints:



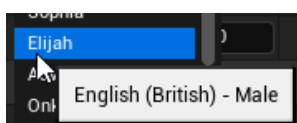
### 4.1. Ariel Text-To-Speech

This function calls the Ariel API using HTTPS request with all defined Ariel parameters (speaker, sentence, format, effects, ...). The response is sent through a delegate :



More information about Ariel function parameters:

**Speaker:** Defines the voice used to play the sentence. Hover the cursor on an item on the list to see more details about it. **Only the voice “Zenaya” is available in the free version!**



⇒ **You can find the list of all speakers at the end of this document!**

**Sentence:** Defines the sentence that the API needs to generate the speech for.

**Audio format:** Defines the format of the audio played. Supported formats are wav, mp3, mp4 and flac.

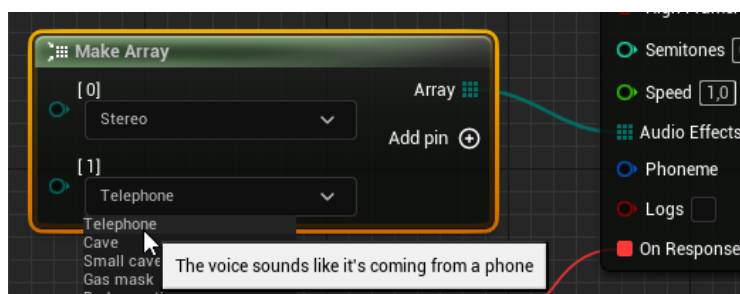
**Volume:** Adjust the volume of the speaker by the specified dB, i.e: +3dB, -5dB.

**High framerate:** Generate a speech using 44.1 KHz frequency instead of 22.05 KHz (default).

**Semitones:** Defines the pitch of the generated speech as semitones (1 octave is +-12st).

**Speed:** Defines the speed generated speech. Normal speed is 1.0.

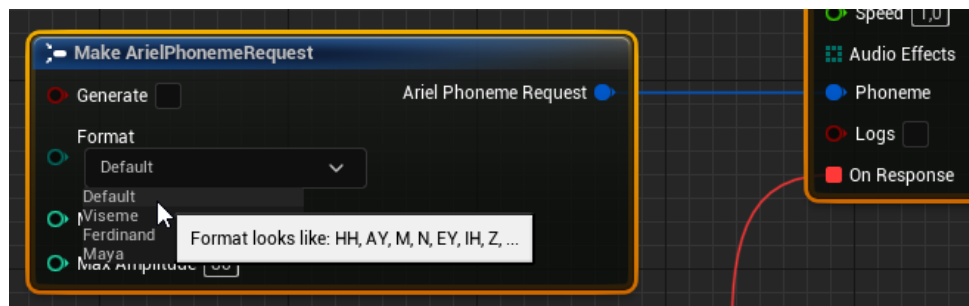
**Audio effects:** Defines all effects who will be applied to the audio. You can combine effects using an array with multiple values. Hover the cursor on an item on the list to see more details about it.



#### Available effects:

- Telephone
- Small cave
- Bad reception
- Stereo
- Cave
- Gas mask
- Next room

**Phonemes:** Indicate how the phonemes need to be generated.



**Generate:** Indicate if the phonemes should be generated or not.

**Format:** Defines the phoneme format. Hover the cursor for more details.

**Min ampl:** Defines the phonemes minimum amplitude (in dB).

**Max ampl:** Defines the phonemes maximum amplitude (in dB).

**Logs:** Indicate if the logs should be printed to the console (Dev only).

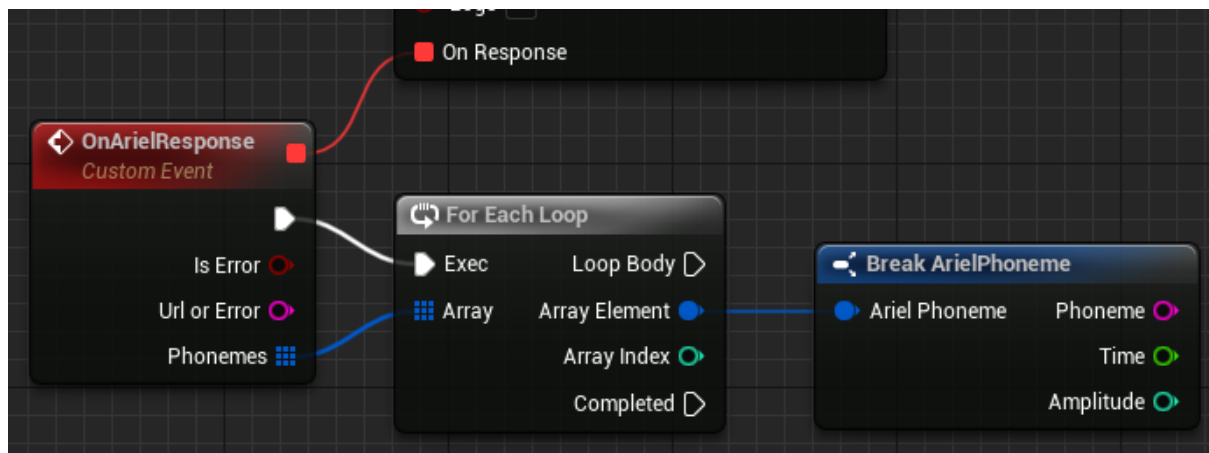
#### More information about Ariel return response:

You can bind and handle Ariel response using the node “Add custom event” or “Create Event”.

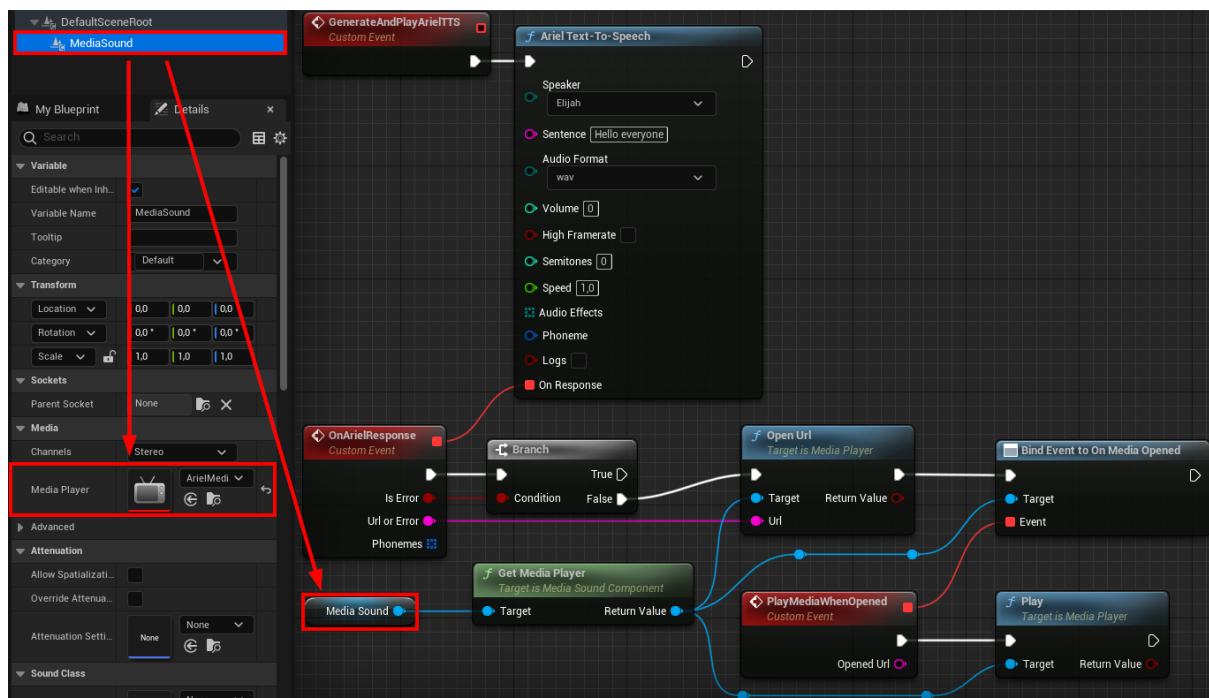
**Is Error:** Indicate if there was an error while processing the request.

**URL or Error:** The generated audio URL (if success) or the error message (if error).

**Phonemes:** *Optional.* The list of all phonemes for the sentence. You may break array elements in order to retrieve the phoneme, the time and the amplitude associated:



The generated audio URL can be passed to an Unreal “Media Player” or “Media steam” in order to be played at runtime. The image below show an example generating and playing the audio at runtime:

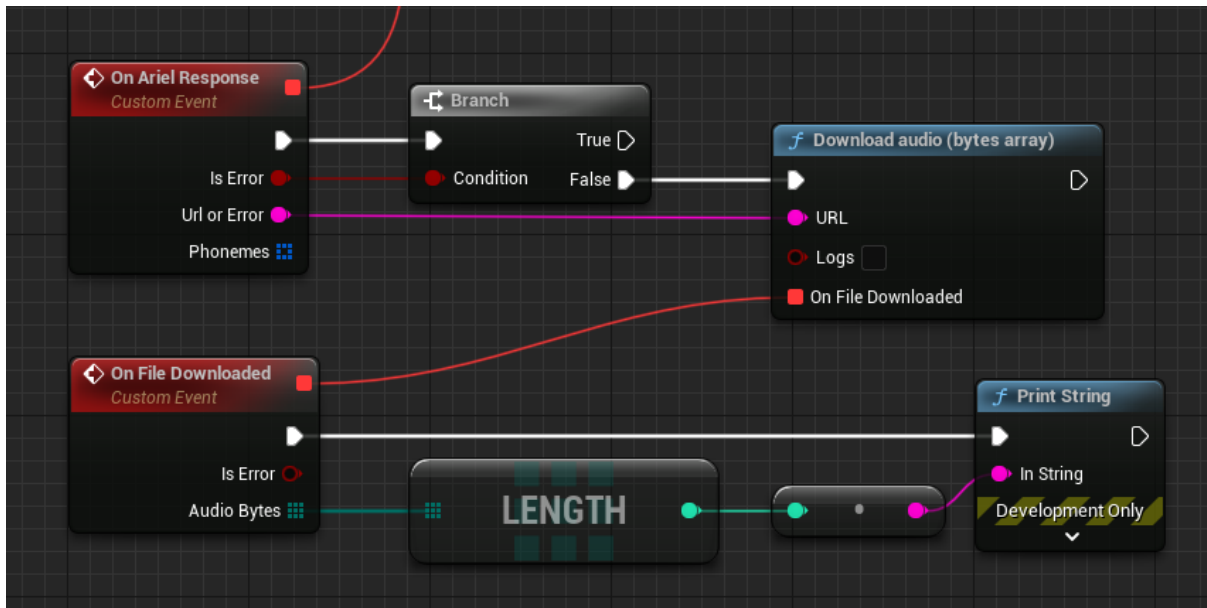


For more informations, please see: <https://blueprintue.com/blueprint/te9qcpy5/>



## 4.2. Download audio as bytes array

This function takes the URL generated from the previous function (*Ariel Text-To-Speech*), downloads the audio and sends it to the event delegate as a byte array:



More information about Download function parameters:

**URL:** The URL of the audio file who needs to be downloaded.

**Logs:** Indicate if the logs should be printed to the console (Dev only).

More information about Download return response:

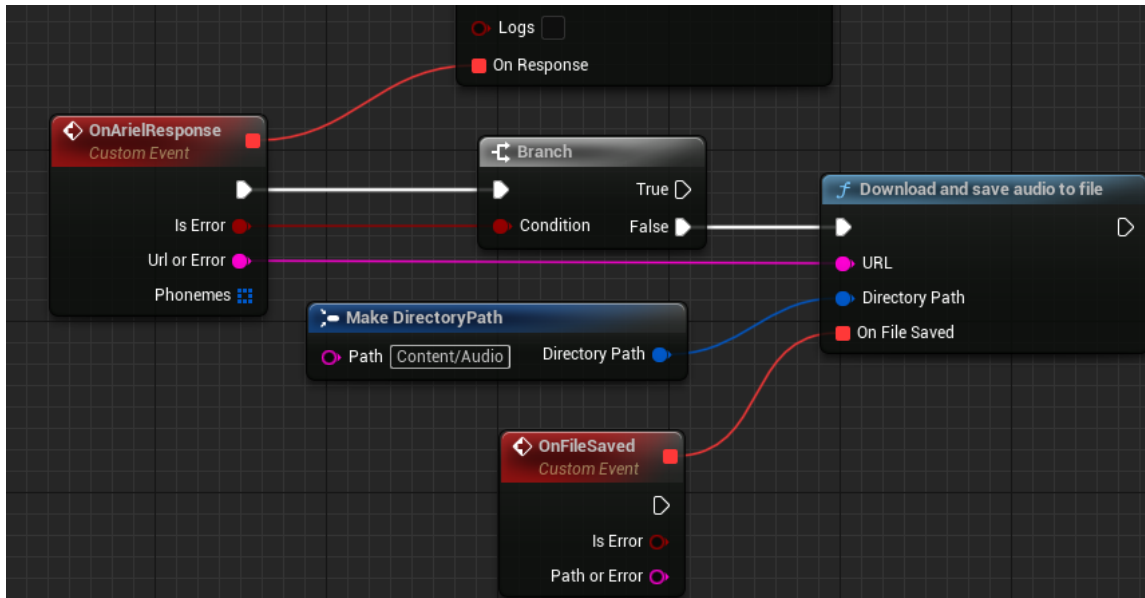
You can bind and handle download response using nodes “Add custom event” or “Create event”.

**Is Error:** Indicate if there was an error while processing the request.

**Audio Bytes:** The audio bytes, stored in an array. (an empty array if there were an error)

### 4.3. Download audio and save it to a file

This function takes the URL generated from the previous function (*Ariel Text-To-Speech*), downloads the audio and stores it in the specified directory. The response is sent through a delegate:



More information about Download function parameters:

**URL:** The URL of the audio file who needs to be downloaded.

**Directory path:** Define the directory where the downloaded file needs to be stored.

More information about Download return response:

You can bind and handle download response using nodes “*Add custom event*” or “*Create event*”.

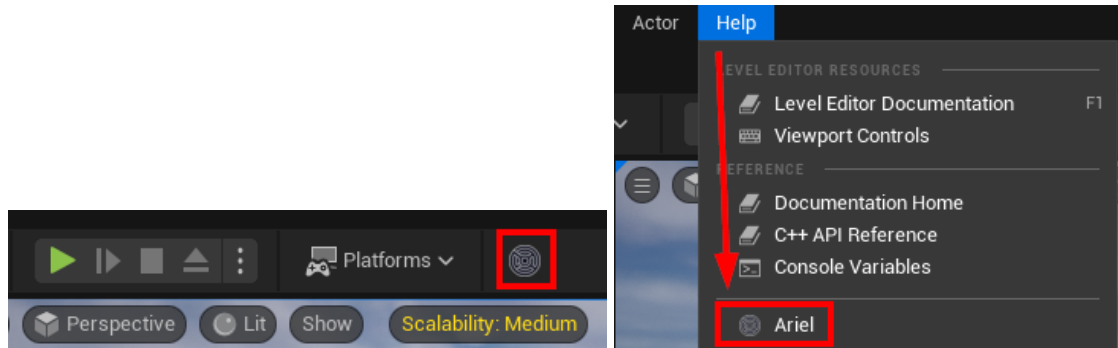
**Is Error:** Indicate if there was an error while processing the request.

**Path or Error:** The audio file location (if successful, with filename) or the error message.

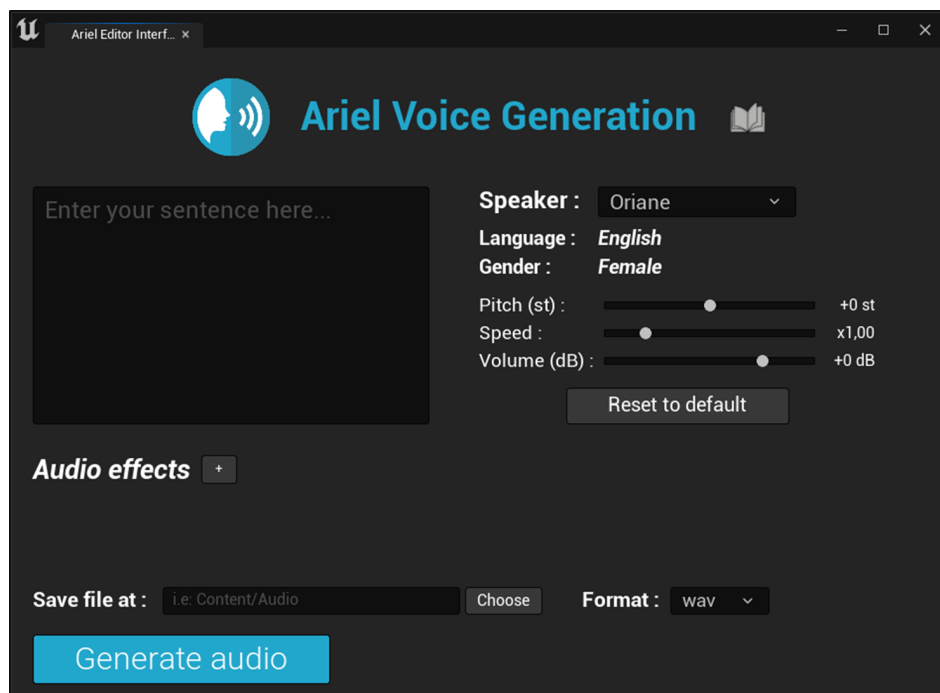
For more informations, please see: <https://blueprintue.com/blueprint/8hvfckk/>

## 5. How to use the Ariel plugin in Editor (with graphical interface)

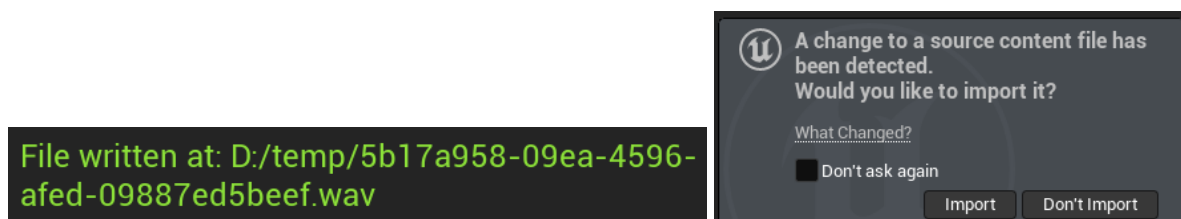
If you want to generate audio sentences that will be saved as a file, you can use the “Editor Utility Widget” included in the plugin. You can open the Window by click the Ariel icon button or in the menu Help > Ariel:



The Ariel Editor Windows should open. Here you can adjust and fill all settings for Ariel. Please read [Ariel Text-To-Speech](#) chapter to find more information about the Ariel parameters.



Then, press the “Generate audio” button and wait for the API to generate and write it into a file. If you enter a path located inside your unreal project, you should see a popup asking you if you want to import the file on the bottom right of the screen:



## 6. Tags

### 6.1. Pause tag

In your text, you can enter a silence tag for a custom pause. Write <pause Xs> or <pause Xms> where X is the duration in seconds or milliseconds. For example "hi how are you? <pause 10s> I am Jane".

→ More tags will be added in the next updates of our plugin !

## 7. Speaker list

This is the list of the Speakers currently supported by Ariel.

### 7.1. Premium voices

| Speaker name | Voice type | Language | Male/Female/Neutral |
|--------------|------------|----------|---------------------|
| Oriane       | Premium    | English  | Female              |
| Dulhan       | Premium    | English  | Male                |
| Alioth       | Premium    | English  | Male                |
| Zenaya       | Premium    | English  | Female              |
| Ada          | Premium    | English  | Female              |
| Socrates     | Premium    | English  | Male                |
| David        | Premium    | English  | Male                |
| Tom          | Premium    | English  | Male                |
| Baldur       | Premium    | English  | Male                |
| Elizabeth    | Premium    | English  | Female              |
| Elen         | Premium    | English  | Female              |
| Riley        | Premium    | English  | Neutral             |
| Orc          | Premium    | English  | Neutral             |
| Urukhai      | Premium    | English  | Neutral             |

|          |         |         |         |
|----------|---------|---------|---------|
| Goblin   | Premium | English | Neutral |
| HighElve | Premium | English | Female  |
| DarkElve | Premium | English | Male    |
| HalfElve | Premium | English | Female  |
| WoodElve | Premium | English | Male    |
| Dis      | Premium | English | Female  |
| Dwarf    | Premium | English | Male    |
| Nordman  | Premium | English | Male    |
| Duergar  | Premium | English | Male    |

## 7.2. Standard voices

| Speaker name | Voice type | Language                | Male/Female |
|--------------|------------|-------------------------|-------------|
| Farah        | Standard   | Arabic                  | Female      |
| Daiyu        | Standard   | Chinese (Mandarin)      | Female      |
| Emma         | Standard   | Danish                  | Female      |
| Oscar        | Standard   | Danish                  | Male        |
| Anke         | Standard   | Dutch                   | Female      |
| Adriaan      | Standard   | Dutch                   | Male        |
| Mia          | Standard   | English (Australian)    | Female      |
| Grace        | Standard   | English (Australian)    | Female      |
| Jack         | Standard   | English (Australian)    | Male        |
| Charlotte    | Standard   | English (British)       | Female      |
| Sophia       | Standard   | English (British)       | Female      |
| Elijah       | Standard   | English (British)       | Male        |
| Advika       | Standard   | English (Indian)        | Female      |
| Onkar        | Standard   | English (Indian)        | Female      |
| Elna         | Standard   | English (South African) | Female      |
| Mary         | Standard   | English (US) - child    | Female      |
| Linda        | Standard   | English (US)            | Female      |
| Patricia     | Standard   | English (US)            | Female      |
| Barbara      | Standard   | English (US)            | Female      |
| Susan        | Standard   | English (US)            | Female      |
| Paul         | Standard   | English (US)            | Male        |
| Michael      | Standard   | English (US) - child    | Male        |
| William      | Standard   | English (US) - child    | Male        |
| Thomas       | Standard   | English (US)            | Male        |

| <b>Speaker name</b> | <b>Voice type</b> | <b>Language</b>        | <b>Male/Female</b> |
|---------------------|-------------------|------------------------|--------------------|
| Aeron               | Standard          | English (Welsh)        | Male               |
| Capucine            | Standard          | French                 | Female             |
| Alix                | Standard          | French                 | Female             |
| Arnaud              | Standard          | French                 | Male               |
| Stephanie           | Standard          | French (Canadian)      | Female             |
| Celine              | Standard          | French (Canadian)      | Female             |
| Maria               | Standard          | German                 | Female             |
| Theresa             | Standard          | German                 | Female             |
| Felix               | Standard          | German                 | Male               |
| Anna                | Standard          | Icelandic              | Female             |
| Sigriour            | Standard          | Icelandic              | Male               |
| Gabriella           | Standard          | Italian                | Female             |
| Bella               | Standard          | Italian                | Female             |
| Lorenzo             | Standard          | Italian                | Male               |
| Rika                | Standard          | Japanese               | Female             |
| Tanaka              | Standard          | Japanese               | Male               |
| Ji-Ho               | Standard          | Korean                 | Female             |
| Camilla             | Standard          | Norwegian              | Female             |
| Katarzyna           | Standard          | Polish                 | Female             |
| Malgorzata          | Standard          | Polish                 | Female             |
| Piotr               | Standard          | Polish                 | Male               |
| Jan                 | Standard          | Polish                 | Male               |
| Tabata              | Standard          | Portuguese (Brazilian) | Female             |
| Juliana             | Standard          | Portuguese (Brazilian) | Female             |
| Pedro               | Standard          | Portuguese (Brazilian) | Male               |

| <b>Speaker name</b> | <b>Voice type</b> | <b>Language</b>       | <b>Male/Female</b> |
|---------------------|-------------------|-----------------------|--------------------|
| Pati                | Standard          | Portuguese (European) | Female             |
| Adriano             | Standard          | Portuguese (European) | Male               |
| Alexandra           | Standard          | Romanian              | Female             |
| Inessa              | Standard          | Russian               | Female             |
| Viktor              | Standard          | Russian               | Male               |
| Francisca           | Standard          | Spanish (European)    | Female             |
| Margarita           | Standard          | Spanish (European)    | Female             |
| Mateo               | Standard          | Spanish (European)    | Male               |
| Leticia             | Standard          | Spanish (Mexican)     | Female             |
| Josefina            | Standard          | Spanish (US)          | Female             |
| Rosa                | Standard          | Spanish (US)          | Female             |
| Miguel              | Standard          | Spanish (US)          | Male               |
| Eva                 | Standard          | Swedish               | Female             |
| Mesut               | Standard          | Turkish               | Female             |
| Angharad            | Standard          | Welsh                 | Female             |